

NISOL INJECTION (10 ML VIAL)

DESCRIPTION: Nisol Injection is a white sterile aqueous suspension.

COMPOSITION: Each ml contains Prednisolone Acetate 25 mg. Preservative: Benzyl Alcohol 10 mg/ml

PHARMACODYNAMICS:

Prednisolone diffuses across cell membranes and complex with specific cytoplasmic receptors. These complexes then enter the cell nucleus, bind to DNA, and stimulate transcription of messenger RNA (mRNA) and subsequent protein synthesis of various enzyme (thought to be ultimately responsible for glucocorticoid effect and slight mineralocorticoid effect). For glucocorticoid effect, particularly anti-inflammatory properties Prednisolone decreases or prevent tissues responses to inflammatory process, thereby reducing development of symptoms of inflammation without effecting the underlying cause. Prednisolone inhibits accumulation of inflammatory cells, including macrophages and leukocytes, at sites of inflammation. They also inhibit phagocytosis, lysosomal enzyme release and synthesis and/or release of several chemical mediators of inflammation.

5 mg of prednisolone is equivalent in anti-inflammatory activity to about 25 mg of cortisone acetate.

PHARMACOKINETICS:

Absorption - Local/Intramuscular: Slowly but completely absorbed.

Distribution - Prednisolone is extensively bound to plasma proteins.

Biotransformation - Primarily hepatic; also renal and tissue; mostly to inactive metabolites.

Elimination - Primarily by renal excretion of inactive metabolites.

INDICATION:

1. Prednisolone Acetate Injection is intended primarily for the local treatment in intra-articular or periarticular injection of rheumatic and other inflammatory conditions. Indications include rheumatoid arthritis and osteoarthritis when few joints are involved.
2. Prednisolone Acetate Injection is suitable for parenteral administration by the intramuscular route in conditions requiring systemic corticosteroids.

RECOMMENDED DOSAGE:

For articular Injection: 5 – 25mg depending upon the size of the joint. The injections may be repeated when relapse occurs. Not more than three joints should be treated in any one day.

For intramuscular Injection: Dosage will depend upon the clinical circumstances and the judgement of the physician. Suggested dose 25 – 100mg once or twice weekly.

Children: Corticosteroids cause growth retardation in infancy, childhood and adolescence, which may be irreversible. Treatment should be limited to the minimum dosage for the shortest time possible.

Elderly: Steroids should be used cautiously in the elderly since adverse effects are enhanced by old age.

Note: When long term treatment is to be discontinued, the dose administered should be gradually reduced over a period of weeks or months depending on dosage and duration of the therapy. Undesirable effects may be minimized by using the lowest effective dose for the minimum period, and by administering the daily requirement as a single morning dose or whenever possible as a single morning dose on alternate days. Frequent patient review is required to appropriate titrate the dose against disease activity

ROUTE OF ADMINISTRATION: For Intra-articular or intramuscular injection

CONTRAINDICATIONS:

For intra-articular injection:

- Arthroplasty of joint, prior-increased risk of infection
- Blood clotting disorders-risk of intra and extra-articular haemorrhage
- Fracture, intra-articular-healing may be retarded
- Infection, periarticular, current or history of-may be exacerbated or reactivated
- Osteoporosis, juxta-articular, non-arthritis-may be exacerbated
- Unstable joint.

For all indications:

Contraindications in patients with known hypersensitivity with any of the ingredients, congestive heart failure, peptic ulcer active tuberculosis, acute bacterial or viral infection and patients undergo vaccination.

WARNINGS & PRECAUTIONS:

Warning:

As this preparation contains benzyl alcohol, its use should be avoided in children under two years of age. Not to be used in neonates.

When the suspension dosage is administered intramuscularly, they should be injected deeply into the gluteal muscle to prevent local tissue atrophy. It is recommended that the deltoid muscle not be used because of a higher incidence of local atrophy. In addition, do not inject repeatedly into the same site. A standard textbook should be consulted for specific techniques and procedures applicable to local injection of corticosteroids for various indications.

It is recommended that intra-articular injections be repeated no more often than once every 3 weeks. Frequent repeated injections may cause joint damage.

Following intra-articular injection, the injected joint should not be overused, even if pain is relieved, because of the increased risk of joint damage or deterioration.

It is recommended that weight-bearing joints be rested for 24 to 48 hours post injection.

Administration of a local anaesthetic concurrently with intra-articular or soft tissue injection of a corticosteroid may reduce the pain of injection and provide immediate relief of symptoms. However, a post-injection flare of pain may occur when the local anaesthetic effect subsides.

Dosages for local injections (e.g., intra-articular, intrabursal, intradermal, intralesional) are given as ranges only. The actual dosage depends upon the size of the joint or lesion and the severity of the condition being treated.

Scleroderma renal crisis

Caution is required in patients with systemic sclerosis because of an increased incidence of (possibly fatal) scleroderma renal crisis with hypertension and decreased urinary output observed with a daily dose of 15 mg or more prednisolone.

Pheochromocytoma crisis, which may be fatal, has been reported after administration of systemic corticosteroids. Corticosteroids should only be administered to patients with suspected or identified pheochromocytoma after an appropriate risk/benefit evaluation

INTERACTION WITH OTHER MEDICAMENTS:

Listed below are some of the possible drug interactions with prednisolone.

Drug	Interacting drugs	Possible effects
Prednisolone	Bendroflumazide, Chlorothiazide, Cyclopenthiazide, Hydrochlorothiazide, Methyclothiazide	The risk of hyperglycaemia caused by prednisolone is increased by the thiazide diuretics.
Prednisolone	Ascorbic acid	Ascorbic acid may protect against corticosteroid induced osteoporosis.
Prednisolone	Ascorbic acid	Ascorbic acid reduces the impairment of polymorphonuclear neutrophils caused by corticosteroids.
Prednisolone	Cyclosporin	Cyclosporin reduces the clearance of prednisolone after prolonged use in transplant patients. Effect only observed after 3-6 months use.
Prednisolone	Rifampicin	Plasma levels and half-life of corticosteroids reduced.
Prednisolone	Rifampicin	Rifampicin reduces plasma levels and increases clearance of prednisolone significantly. Suggested that dose of Prednisolone be doubled if rifampicin given.

Co-treatment with CYP3A inhibitors, including cobicistat-containing products, is expected to increase the risk of systemic side-effects. The combination should be avoided unless the benefit outweighs the increased risk of systemic corticosteroid side-effects, in which case patients should be monitored for systemic corticosteroid side-effects.

USE IN PREGNANCY & LACTATION:

Use in Pregnancy: Animal reproduction studies have shown an adverse effect on the foetus and there are no adequate and well controlled studies in humans, but potential benefits may warrant use of the drugs in pregnant women despite potential risks.

Administration of low pharmacologic doses (the equivalent or less of 5 mg of prednisolone per day) is not considered likely to affect the infant adversely

Use in Lactation: Less than 1% of the administered dose of prednisolone is excreted in breast milk. However, breast-feeding during the use of higher pharmacologic doses is not recommended because corticosteroids are excreted in breast milk and may cause unwanted effects, such as growth suppression and inhibition of endogenous steroid production, in the infant.

SIDE EFFECTS:

With intra-articular and other local injections, the principal adverse effect encountered is a temporary local exacerbation with increased pain and swelling. Normally, this subsides after a few hours. Particularly after high prolonged local administration corticosteroids can be absorbed in amounts sufficient to produce systemic effects.

Note: When being given by intramuscular injection, the precautions, adverse effects, contra-indications and withdrawal symptoms associated with the systemic use of corticoids become relevant to Prednisolone Acetate Injection.

Increased joint pain-following intra-articular injection.

SYMPTOMS AND TREATMENT OF OVERDOSE:

Treatment: There is no specific antidote. Toxic effects are signs of over dosage, and should be treated symptomatically and dosage reduced or the drug withdrawn. During long courses of treatment laboratory and metabolic studies should be made. Fluid retention should be watched for via a fluid balance chart and daily weighing. Sodium intake may need to be reduced to less than 1g daily and potassium supplements may be necessary.

INCOMPATIBILITIES: Prednisolone Acetate injection is incompatible with alkalis.

STORAGE CONDITIONS:

Store below 30°C. Protect from light. Protect from freezing. Shake well.
KEEP OUT REACH OF CHILDREN. JAUHKAN DARIPADA KANAK-KANAK

SHELF LIFE: Please refer to outer package.

PACK SIZE: Pack in box of 25 x 10 ml vials

PRODUCT REGISTRATION HOLDER & MANUFACTURER:

DUOPHARMA (M) SDN BHD
Lot 2599, Jalan Seruling 59 Kawasan 3,
Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA.